Code No: 135CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Define the term modulation and give the classification of analog modulation techniques. [2]
b) Define the terms Channel Bandwidth and Signal Bandwidth. [3]
c) Define the term Frequency Modulation. [2]
d) List out the advantages and disadvantages of Digital Modulation systems over analog modulation systems. [3]
e) Define Internet Telephony [2]
f) Discuss about various types of paging systems. [3]
g) How light travels through fiber optics? [2]
h) Write about the characteristics of optical fiber transmitter elements. [3]
i) Explain WCDMA technology. [2]
j) Write about GSM technology. [3]

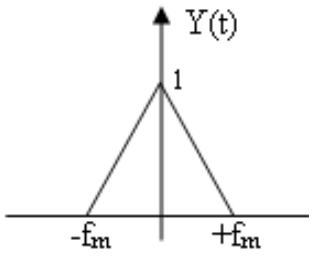
PART – B

(50 Marks)

2.a) Explain the need for modulation in communication systems.
b) Explain the significance of systems gain and attenuation of signals in communication systems. [6+4]

OR

3.a) Draw the electromagnetic spectrum and specify its applications.
b) Draw the spectrum representation of the signal $x_1(t)=A*Cos(2*pi*f*t)$ and $x_2(t) = y(t)*Cos(2*pi*f*t)$. Given y(t) as below. [5+5]



4.a) Represent the data stream 1 0 1 1 0 0 0 1 with BFSK and BPSK modulation schemes.
b) Illustrate with necessary diagrams the working of Quadrature Phase Shift Keying (QPSK). [4+6]

OR

5.a) Explain with necessary diagrams the generation of Pulse Amplitude Modulated (PAM) Signal.
b) Compare Amplitude Modulation and Frequency Modulation Systems. [6+4]

6.a) Compare the networks LAN, PAN and WAN.
b) Explain with required diagrams the working of PSTN. [4+6]

OR

7.a) Write about Token Ring LAN architecture.
b) Illustrate with necessary diagrams various types of Bus topology. [5+5]

8.a) Explain the principal of trilateration.
b) Write about various satellite Subsystems. [5+5]

OR

9.a) What are the advantages, applications and disadvantages of optical fiber as compared to copper cables?
b) Explain the role of satellite ground stations, telemetry and tracking systems. [4+6]

10. List out the types of multiple access Systems. Explain the working of CDMA scheme. [10]

OR

11.a) Write the applications of RFID communication.
b) Discuss about Mesh Wireless networks. [4+6]

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